

CLOUDFORMATION

AWS CloudFormation is a service in Amazon Web Services that allows you to create and manage AWS resources using code (templates) YAML or JSON templates instead of manual setup.

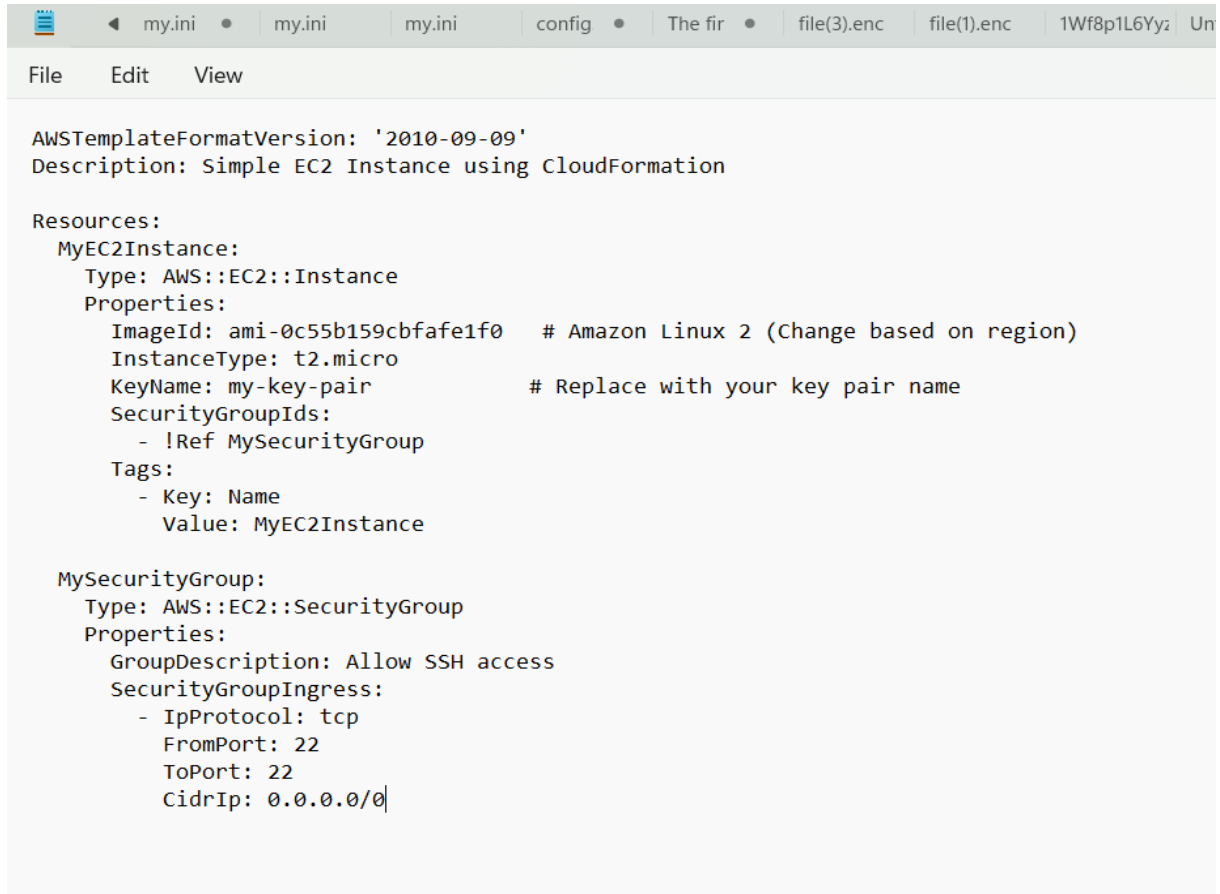
Features

- Infrastructure as Code (IaC)
- Automation
- Reusable templates
- Easy updates and rollback
- Supports multiple AWS services

HANDS-ON FOR CLOUD FORMATION

Screenshot: Save CloudFormation Template

- Create EC2 creation template in YAML format
- Save the file in Notepad
- Ensure the file is saved with .yaml extension



The screenshot shows a Notepad application window with a menu bar (File, Edit, View) and a tab bar containing several files: my.ini, my.ini, my.ini, config, The fir, file(3).enc, file(1).enc, 1Wf8p1L6Yyz, and Un. The main text area displays a CloudFormation template in YAML format. The template defines an EC2 instance and a security group. The EC2 instance is named 'MyEC2Instance' and is of type 'AWS::EC2::Instance'. It has properties for 'ImageId' (ami-0c55b159cbfafa1f0), 'InstanceType' (t2.micro), 'KeyName' (my-key-pair), and 'SecurityGroupIds' (referring to 'MySecurityGroup'). It also has a 'Tags' property with a key 'Name' and a value 'MyEC2Instance'. The security group is named 'MySecurityGroup' and is of type 'AWS::EC2::SecurityGroup'. It has a 'GroupDescription' of 'Allow SSH access' and a 'SecurityGroupIngress' property with a rule for 'IpProtocol: tcp', 'FromPort: 22', 'ToPort: 22', and 'CidrIp: 0.0.0.0/0'.

```
AWSTemplateFormatVersion: '2010-09-09'
Description: Simple EC2 Instance using CloudFormation

Resources:
  MyEC2Instance:
    Type: AWS::EC2::Instance
    Properties:
      ImageId: ami-0c55b159cbfafa1f0    # Amazon Linux 2 (Change based on region)
      InstanceType: t2.micro
      KeyName: my-key-pair              # Replace with your key pair name
      SecurityGroupIds:
        - !Ref MySecurityGroup
      Tags:
        - Key: Name
          Value: MyEC2Instance

  MySecurityGroup:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: Allow SSH access
      SecurityGroupIngress:
        - IpProtocol: tcp
          FromPort: 22
          ToPort: 22
          CidrIp: 0.0.0.0/0
```

Screenshot: Create Key Pair

- Go to EC2 Dashboard
- Click on Key Pairs
- Click on Create key pair
- Enter key pair name
- Download the key pair file

The screenshot shows the 'Create key pair' page in the AWS Management Console. The breadcrumb navigation at the top reads 'EC2 > Key pairs > Create key pair'. The main heading is 'Create key pair' with an 'Info' link. Below this, a description states: 'A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.' The 'Name' field contains 'my-key-pair' with a note: 'The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.' The 'Key pair type' section has two radio buttons: 'RSA' (selected) and 'ED25519'. The 'Private key file format' section has two radio buttons: '.pem' (selected, with the note 'For use with OpenSSH') and '.ppk' (with the note 'For use with PuTTY'). A 'Tags - optional' section indicates 'No tags associated with the resource' and includes an 'Add new tag' button with the note 'You can add up to 50 more tags.' At the bottom right are 'Cancel' and 'Create key pair' buttons.

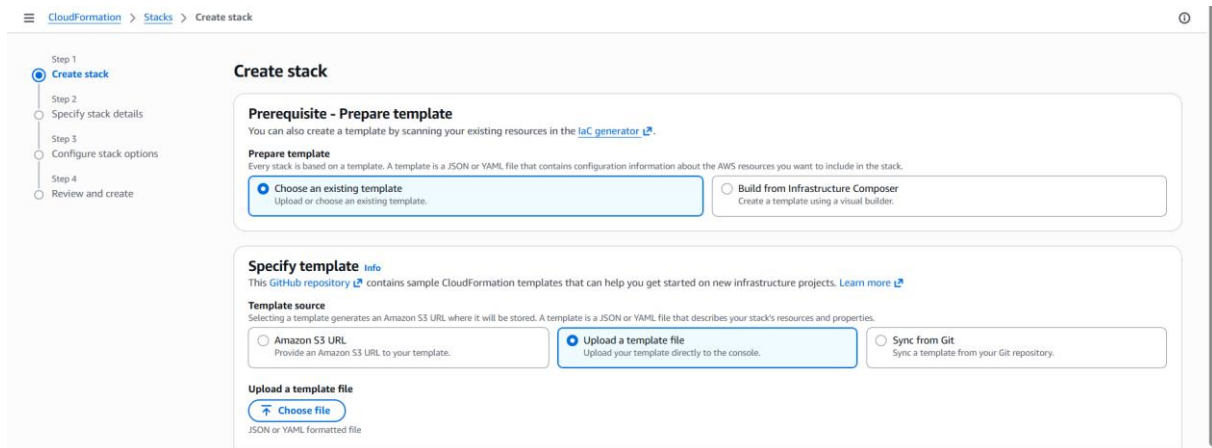
Screenshot: Create Stack

- Go to CloudFormation Dashboard
- Click on Create stack

The screenshot shows the 'Stacks' page in the AWS CloudFormation console. The breadcrumb navigation is 'CloudFormation > Stacks'. The left sidebar contains links for 'Stacks', 'Stack refactors', 'StackSets', 'Exports', 'Infrastructure Composer', 'IaC generator', 'Hooks overview', 'Invocation summary', and 'Hooks'. The main area is titled 'Stacks (0)' and includes a search bar 'Search by stack name'. Above the table are buttons for 'Delete stack', 'Update stack', 'Stack actions', and 'Create stack'. A 'Filter status' dropdown is set to 'Active', and a 'View nested' toggle is turned on. The table has columns for 'Stack name', 'Status', 'Created time', and 'Description'. Below the table, a message states 'No stacks' and 'No stacks to display', with a 'Create stack' button and a 'View getting started guide' link.

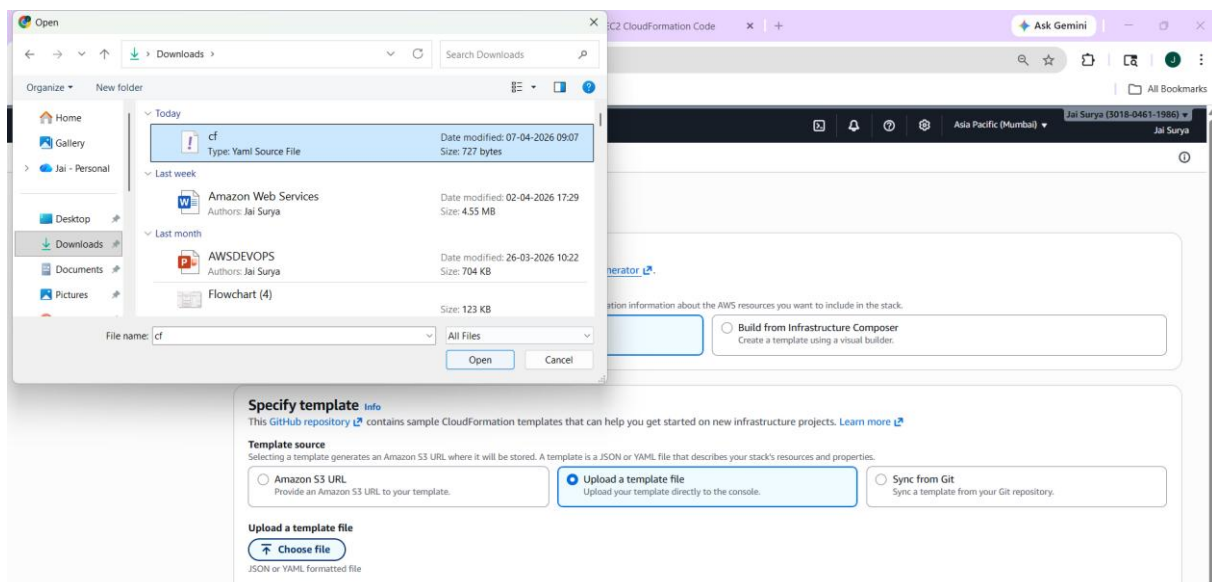
Screenshot: Upload Template

- Select Choose an existing template
- Select Upload a template file



Screenshot: Select Template File

- Click on Choose file
- Select the saved YAML template file



Screenshot: Configure Stack Details

- Enter stack name
- Provide parameter values

0

The screenshot shows the 'Specify stack details' step in the AWS CloudFormation console. On the left, a progress bar indicates the steps: Step 1: Create stack, Step 2: Specify stack details (selected), Step 3: Configure stack options, and Step 4: Review and create. The main area is titled 'Specify stack details' and contains two sections: 'Provide a stack name' and 'Parameters'. In the 'Provide a stack name' section, the 'Stack name' field is filled with 'mystack'. Below it, a note states: 'Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 7/128.' The 'Parameters' section has a sub-header 'Parameters' and a note: 'Parameters are defined in your template and allow you to input custom values when you create or update a stack.' It contains two parameter fields: 'InstanceType' with the value 't3.micro' and 'KeyPairName' with a dropdown menu showing '3pmibatch2' selected. The dropdown menu also shows 'KeyName' as an option.

Screenshot: Stack Creation Completed

- Verify that the stack is created successfully
- Ensure the stack status shows CREATE_COMPLETE

The screenshot shows the AWS CloudFormation console with the 'mystack' stack selected. The left sidebar shows the 'Stacks (1)' list with a search bar and a filter set to 'Active'. The 'mystack' stack is listed with a status of 'CREATE_COMPLETE' and a creation time of '2026-04-07 09:19:51 UTC+0530'. The main area shows the 'mystack' stack details, including a 'Stack info' tab and a 'Stack actions' menu. The 'Overview' section displays the 'Stack ID' as 'arn:aws:cloudformation:ap-south-1:301804611986:stack/mystack/44d43040-3234-11f1-a66a-068f550d5f11'. The 'Status' is 'CREATE_COMPLETE'. The 'Status reason' is '-'. The 'Parent stack' is '-'. The 'Created time' is '2026-04-07 09:19:51 UTC+0530'. The 'Updated time' is '-'. The 'Deleted time' is '-'. The 'Last drift check time' is '-'. The 'Drift status' is 'NOT_CHECKED'. The 'Termination protection' is 'Deactivated'.

Screenshot: EC2 Instance Created Successfully

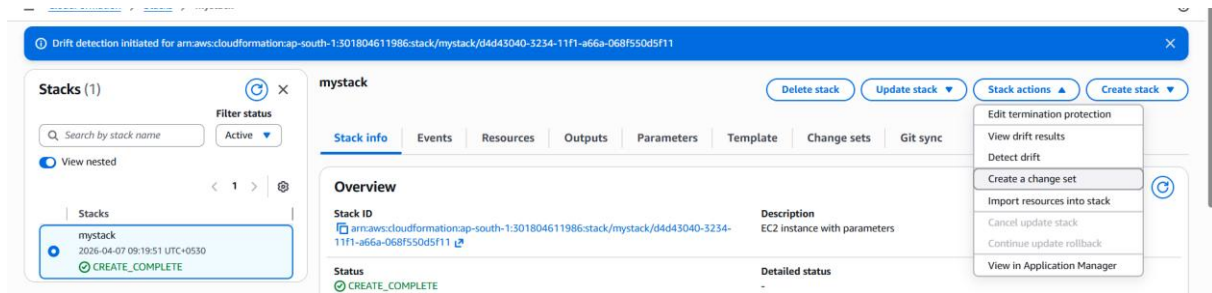
- Verify that the EC2 instance is created successfully
- Ensure the instance is in running state

The screenshot shows the AWS Management Console with the 'EC2' service selected. The left sidebar shows the 'Instances (1)' list with a search bar and a filter set to 'Active'. The 'Param-EC2' instance is listed with a status of 'Running' and a creation time of '2026-04-07 09:19:51 UTC+0530'. The main area shows the 'Param-EC2' instance details, including a 'Connect' button, an 'Instance state' dropdown, and an 'Actions' menu. The 'Overview' section displays the 'Instance ID' as 'i-069d513f6706289d3'. The 'Instance state' is 'Running'. The 'Instance type' is 't3.micro'. The 'Status check' is 'Initializing'. The 'Alarm status' is 'View alarms +'. The 'Availability Zone' is 'ap-south-1a'. The 'Public IPv4 DNS' is 'ec2-43-205-233-126.ap...'. The 'Public' checkbox is checked.

Screenshot: Detect Stack Drift

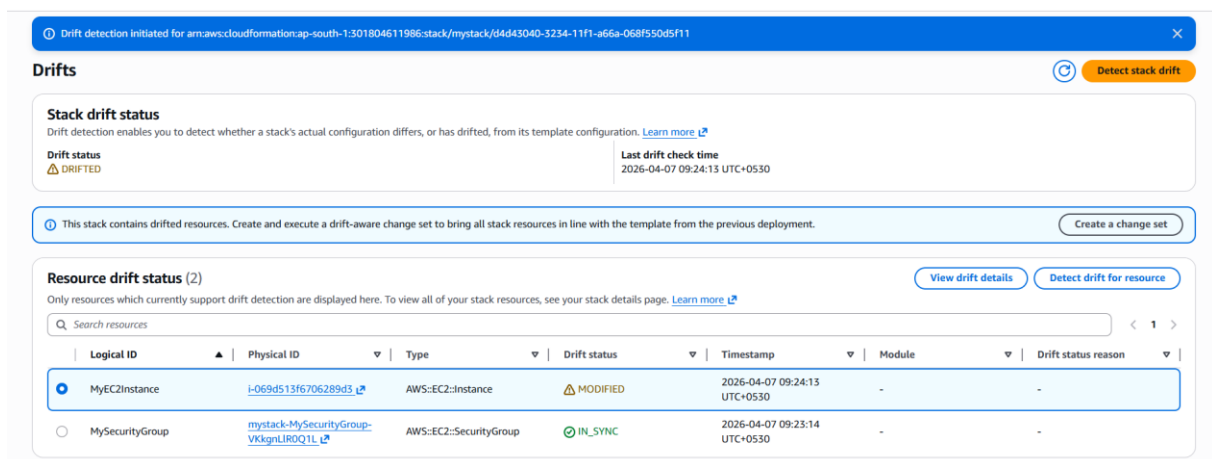
- Go to CloudFormation stack

- Click on Stack actions
- Select Detect drift



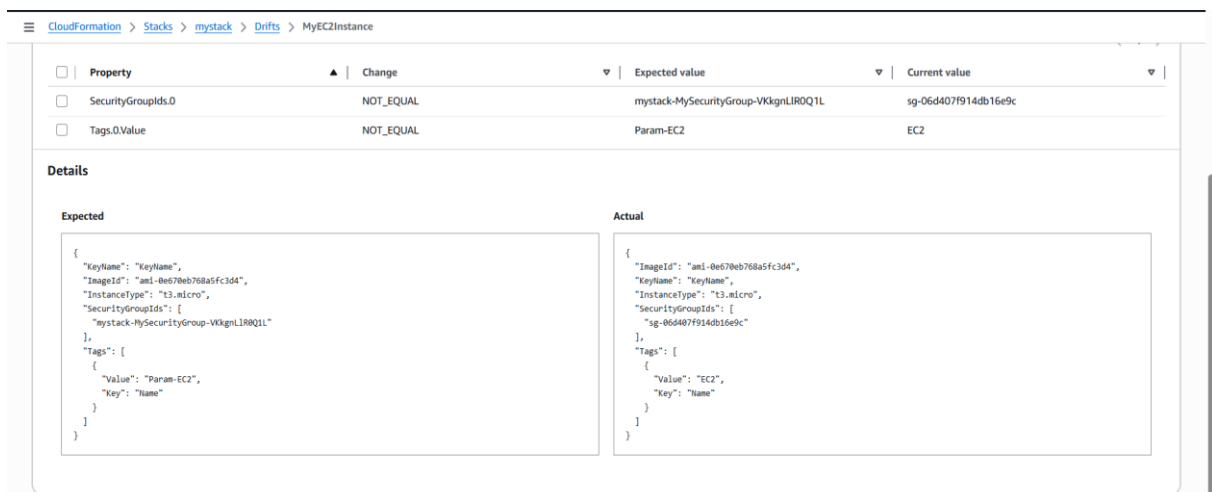
Screenshot: View Drift Details

- Select the created instance resource
- Click on View drift details



Screenshot: Drift Details Comparison

- View expected configuration details
- View actual configuration details
- Compare both to identify drift



TROUBLE SHOOTING COMMANDS

uptime

```
[ec2-user@ip-172-31-40-143 ~]$ uptime
11:35:24 up 2 min,  1 user,  load average: 0.03, 0.06, 0.02
[ec2-user@ip-172-31-40-143 ~]$
```

Check system load

top

```
top - 11:35:52 up 3 min,  1 user,  load average: 0.02, 0.05, 0.02
tasks: 108 total,  1 running, 107 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  916.8 total,  560.4 free,  128.1 used,  228.3 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  652.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 107612  17712  11048 S   0.0   1.9   0:00.86 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 slab_flushwq
    6 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 netns
    7 root        20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0-cgroup_bpf_destroy
    8 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
    9 root        20   0      0      0      0 I   0.0   0.0   0:00.02 kworker/u4:0-events_unbound
   10 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   11 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   12 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   13 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   14 root        20   0      0      0      0 S   0.0   0.0   0:00.03 ksoftirqd/0
   15 root        20   0      0      0      0 T   0.0   0.0   0:00.01 rcu_preempt
```

CPU & RAM usage

free -m

```
[ec2-user@ip-172-31-40-143 ~]$ free -m
              total        used        free      shared  buff/cache   available
Mem:           916          128          560           0          228          652
Swap:           0           0           0
```

Disk Issues

df -h

```
[ec2-user@ip-172-31-40-143 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0    4.0M   0% /dev
tmpfs           459M   0    459M   0% /dev/shm
tmpfs           184M 440K   183M   1% /run
/dev/nvme0n1p1  8.0G  1.6G   6.4G  20% /
tmpfs           459M   0    459M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
tmpfs           92M   0     92M   0% /run/user/1000
[ec2-user@ip-172-31-40-143 ~]$
```

Website Not Working (HTTP Issue)

netstat -tulnp | grep 80

```
[ec2-user@ip-172-31-40-143 ~]$  
[ec2-user@ip-172-31-40-143 ~]$ netstat -tulnp | grep 80  
(Not all processes could be identified, non-owned process info  
will not be shown, you would have to be root to see it all.)  
udp6      0      0 fe80::c3:7ff:fe3c:e:546 :::*  
[ec2-user@ip-172-31-40-143 ~]$
```

Logs (MOST IMPORTANT IN EC2)

journalctl -xe

```
[ec2-user@ip-172-31-40-143 ~]$  
[ec2-user@ip-172-31-40-143 ~]$ journalctl -xe  
  
A start job for unit refresh-policy-routes@ens5.service has begun execution.  
  
The job identifier is 971.  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: Starting configuration refresh for ens5  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: Got IMDSv2 token for interface ens5 from http://169.254.169.254/latest via ens5  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: [get_meta] Querying IMDS for mac  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: [get_meta] Querying IMDS for mac  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: Using existing cfgfile /run/systemd/network/70-ens5.network  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: [get_meta] Querying IMDS for mac  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: [get_meta] Querying IMDS for mac  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: [get_meta] Querying IMDS for network/interfaces/mac802:3:07:3c:e9:fb/local-ipv4s  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: Called trap  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal ec2net[2094]: No networkd reload needed  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal systemd[1]: refresh-policy-routes@ens5.service: Deactivated successfully.  
Subject: Unit succeeded  
Defined-By: systemd  
Support: https://lists.freedesktop.org/mailman/listinfo/systemd-devel  
  
The unit refresh-policy-routes@ens5.service has successfully entered the 'dead' state.  
Apr 07 11:36:22 ip-172-31-40-143.ap-south-1.compute.internal systemd[1]: Finished refresh-policy-routes@ens5.service - Refresh policy routes for ens5.  
Subject: A start job for unit refresh-policy-routes@ens5.service has finished successfully  
Defined-By: systemd  
Support: https://lists.freedesktop.org/mailman/listinfo/systemd-devel  
  
A start job for unit refresh-policy-routes@ens5.service has finished successfully.  
  
The job identifier is 971.
```

Service Troubleshooting

systemctl status sshd

```
[ec2-user@ip-172-31-40-143 ~]$  
[ec2-user@ip-172-31-40-143 ~]$ systemctl status sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)  
   Active: active (running) since Tue 2026-04-07 11:32:35 UTC; 6min ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 1587 (sshd)  
    Tasks: 1 (limit: 1014)  
  Memory: 7.6M  
     CPU: 882ms  
  CGroup: /system.slice/ssh.service  
          └─1587 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Apr 07 11:32:35 ip-172-31-40-143.ap-south-1.compute.internal systemd[1]: Starting sshd.service - OpenSSH server daemon...  
Apr 07 11:32:35 ip-172-31-40-143.ap-south-1.compute.internal sshd[1587]: Server listening on 0.0.0.0 port 22.  
Apr 07 11:32:35 ip-172-31-40-143.ap-south-1.compute.internal sshd[1587]: Server listening on :: port 22.  
Apr 07 11:32:35 ip-172-31-40-143.ap-south-1.compute.internal systemd[1]: Started sshd.service - OpenSSH server daemon.  
Apr 07 11:33:04 ip-172-31-40-143.ap-south-1.compute.internal ec2-instance-connect[1768]: Querying EC2 Instance Connect keys for matching fingerprint:  
Apr 07 11:33:05 ip-172-31-40-143.ap-south-1.compute.internal ec2-instance-connect[1958]: Querying EC2 Instance Connect keys for matching fingerprint:  
Apr 07 11:33:05 ip-172-31-40-143.ap-south-1.compute.internal ec2-instance-connect[1986]: Providing ssh key from EC2 Instance Connect with fingerprint  
Apr 07 11:33:05 ip-172-31-40-143.ap-south-1.compute.internal sshd[1662]: Accepted publickey for ec2-user from 13.233.177.4 port 19932 ssh2: ED25519 S  
Apr 07 11:33:05 ip-172-31-40-143.ap-south-1.compute.internal sshd[1662]: pam_unix(sshd:session): session opened for user ec2-user(uid=1000) by (uid=0  
lines 1-21/21 (END)
```

systemctl list-units --type=service

```
[ec2-user@ip-172-31-40-143 ~]$  
[ec2-user@ip-172-31-40-143 ~]$ systemctl list-units --type=service  


| UNIT                       | LOAD   | ACTIVE | SUB     | DESCRIPTION                                       |
|----------------------------|--------|--------|---------|---------------------------------------------------|
| acpid.service              | loaded | active | running | ACPI Event Daemon                                 |
| amazon-ssm-agent.service   | loaded | active | running | amazon-ssm-agent                                  |
| atd.service                | loaded | active | running | Deferred execution scheduler                      |
| auditd.service             | loaded | active | running | Security Auditing Service                         |
| chronyd.service            | loaded | active | running | NTP client/server                                 |
| cloud-config.service       | loaded | active | exited  | Apply the settings specified in cloud-config      |
| cloud-final.service        | loaded | active | exited  | Execute cloud user/final scripts                  |
| cloud-init-local.service   | loaded | active | exited  | Initial cloud-init job (pre-networking)           |
| cloud-init.service         | loaded | active | exited  | Initial cloud-init job (metadata service crawler) |
| dbus-broker.service        | loaded | active | running | D-Bus System Message Bus                          |
| dracut-shutdown.service    | loaded | active | exited  | Restore /run/initramfs on shutdown                |
| getty@tty1.service         | loaded | active | running | Getty on tty1                                     |
| gssproxy.service           | loaded | active | running | GSSAPI Proxy Daemon                               |
| irqbalance.service         | loaded | active | running | irqbalance daemon                                 |
| kmod-static-nodes.service  | loaded | active | exited  | Create List of Static Device Nodes                |
| libstoragemgmt.service     | loaded | active | running | libstoragemgmt plug-in server daemon              |
| policy-routes@ens5.service | loaded | active | exited  | Set up policy routes for ens5                     |
| rpc-statd-notify.service   | loaded | active | exited  | Notify NFS peers of a restart                     |


```